Mert Tereci

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Military Service: Deferred until December 2028



EDUCATION

ONDOKUZ MAYIS UNIVERSITY — Samsun, Türkiye
B.Sc. in Computer Engineering, GPA 3.16/4.00 — *Jan 2023 - Oct 2026*

BILKENT UNIVERSITY — Ankara, Türkiye
B.Sc. in Computer Engineering, transferred to Ondokuz Mayıs University — *Sep 2018 - Jan 2023*
English Preparatory Program.

Achievement: Ranked 4,501st in the 2018 national university entrance exam (YKS, STEM track)

EXPERIENCE

WEBOFISI YAZILIM HIZMETLERI TIC. LTD. STI. — Samsun, Türkiye
Mobile Application Developer Intern | Flutter, Dart, Provider, REST API — *Jul 2025 - Aug 2025*

- Built a Flutter e-commerce app for selling web scripts and hosting packages as the sole developer during a 30-day internship, including a 158-product XML catalog with search, category filtering and pagination.
- Set up a json-server mock that mirrored the real database schema while the company's backend API was still unavailable, so switching to the production API would mostly require changing the server URL.
- Used Provider for state management and separated screens from business logic and data services in an MVVM-like layered structure.
- Implemented the account flow, including multi-step registration with validation, persistent sessions, profile management and per-user favorites.

PROJECTS

Netflix IMDb Rating Extension | Python, FastAPI, PostgreSQL, Redis, JavaScript, Pandas

- Developed a full-stack Chrome extension that shows IMDb ratings on Netflix title cards, backed by a custom database and REST API instead of a third-party rating service.
- Built a Pandas ETL pipeline that processed over 10 million rows of official IMDb data, filtered out titles with fewer than 1,000 votes and loaded about 60K movies and series into PostgreSQL, along with nearly 24K Turkish titles for local search.
- Designed a three-layer cache (browser, Redis, PostgreSQL) to keep repeated lookups away from the database. In production, Redis hits returned about 40% faster than database queries.
- Replaced per-card requests with batched lookups of up to 20 visible titles, cutting the number of requests by up to 20x.
- Deployed the system on Vercel, Supabase and Upstash Redis, and documented key architecture decisions as ADRs.

2048 AI Player | Python, Pygame, NumPy, Numba

- Built 2048 in Pygame along with an AI that plays the game on its own. At each move, the AI simulates possible future states and scores the board with a custom evaluation function combining 6 criteria such as tile ordering, empty space and merge opportunities.
- Compared Alpha-Beta pruning, which treats the game as if played against an opponent, with Expectimax, which accounts for randomly spawning tiles. Expectimax performed far better, showing that modeling the problem correctly mattered more than adding computation.
- Sped up the computation with Numba, which made deeper search practical and let the AI reach the 2048 tile in nearly every game.

Deepfake Detection with Deep Learning (Senior Thesis, team of three) | Python, PyTorch, OpenCV, ViT, CNN

- Trained and compared 9 CNN and Transformer models under identical conditions to detect AI-generated face videos, showing that Transformer-based models were more consistent.
- Built a preprocessing pipeline that samples frames at even intervals and detects and crops faces, producing a training set of about 375K face images from FaceForensics++ and Celeb-DF v2.
- Created balanced dataset variants because fake samples outnumbered real ones 5 to 6 times, and showed that one model reached 84% accuracy while recognizing only 10% of real images. This made class-based metrics the basis of our evaluation.
- Developed CutMixup, a data augmentation method that mixes training images to discourage memorization, and applied it to ViT-Base. Real-image recall rose from 83% to 93%, and AUC on Celeb-DF v2 reached 99.89%, on par with the best results in the literature.

OHS Tracking App (İSG Takip) | Flutter, Dart, Provider

- Developed a Flutter app that lets an occupational health and safety specialist manage documents and employee records for multiple client companies in one place. The app was used in real day-to-day work.
- Gathered requirements directly from the user and, based on feedback during development, added dated per-employee tracking of training attendance and pre-employment medical exams.
- Assigned a standard OHS document set to each new company automatically and summarized document status and missing employee records. Managed state with Provider and stored data locally on the device.

SKILLS

Programming Languages: Java, Python, SQL, JavaScript, Dart
Frameworks, Libraries & Tools: Flutter, Provider, PyTorch, Pygame, FastAPI, NumPy, Numba, Pandas, Git
Databases: PostgreSQL, Redis, Supabase
Concepts: OOP, REST API Design, Clean Architecture, MVVM
Languages: Turkish (Native), English (Advanced, Bilkent Prep, EF Oxford course 2016), German (Elementary)
Certifications: Flutter & Dart, Udemy (Jul 2023)